AYUSH RAKESHBHAI POSHIYA

Computer Engineering Student • Data Science & AI/ML
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SUMMARY

Second-year B.Tech Computer Engineering student at CHARUSAT University focused on building machine learning applications, analyzing data and developing intelligent solutions. Strong foundation in Data Structures & Algorithms (C++, Python; 150+ problems solved), with hands-on experience across the complete data loop: exploratory data analysis, model training and evaluation, and packaging models into responsive applications using RAG, LLM APIs and FastAPI. Seeking a Data Science / AI-ML internship.

TECHNICAL SKILLS

Programming	Python, C++, Java, SQL
Web Development	HTML, CSS, JavaScript
Data Science	NumPy, Pandas, Matplotlib
AI / Machine Learning	Scikit-learn, NLP, TF-IDF, Cosine Similarity
AI Application Development	FastAPI, RAG, LLM APIs
Tools & Platforms	Git, GitHub, Jupyter, VS Code

PROJECTS

Mission Readiness & Predictive Maintenance Copilot | *AI Maintenance Platform*

Tech: Python, FastAPI, React, Scikit-learn, Pandas, NumPy, SQLite, WebSockets, Groq Llama 3.3 70B, Gemini, Docker

- Built a full-stack platform integrating three ML models (AI4I 2020, NASA IMS Bearing, NASA N-CMAPSS) to predict failure risk, operational readiness and remaining useful life (RUL).
- Implemented an Explainable AI engine using counterfactual feature analysis to identify sensor-level contributors to failure risk across bearing, ground-armor and turbofan systems.
- Added real-time WebSocket telemetry streaming and vectorized batch inference for fleet monitoring, with anomaly/stress simulation and live readiness updates.
- Developed a bilingual English/Hinglish AI Copilot (Groq Llama 3.3 70B with Gemini fallback, local RAG, agentic tools) and a React + SQLite dashboard that generates maintenance work orders.

Cortex | *Agentic RAG Research Assistant*

Tech: Python, FastAPI, Streamlit, ChromaDB, Sentence-Transformers, Gemini, OpenAI, Groq, Docker, Pytest

- Built a full-stack RAG assistant for semantic question-answering over PDF, DOCX, TXT and CSV files, with grounded answers and document/page-level source citations.
- Developed the ingestion and search pipeline: structure-aware chunking, metadata extraction, deduplication, Sentence-Transformers embeddings and cosine-similarity retrieval on ChromaDB.
- Implemented agentic retrieval with query-intent classification, sub-query decomposition, custom lexical-semantic reranking and iterative query refinement using relevance thresholds.
- Designed a provider-agnostic LLM layer (Gemini, OpenAI, Groq); created a Pytest evaluation suite (Precision@K, Recall@K, MRR, Hit Rate@K, faithfulness) and containerized with Docker.

EDUCATION

B.Tech, Computer Engineering – CHARUSAT University **Expected 2029**

1st Year CGPA: 7.80/10 | Class XII: 85.50% | Class X: 86.33%

ACHIEVEMENTS

- Solved 150+ LeetCode problems covering Arrays, Linked Lists, Stacks, Queues, Trees and Binary Search.
- Participated in 5 hackathons, including IBM BOB.

LEADERSHIP & EXTRACURRICULAR

- NSS Volunteer: village cleanliness drives and 7 days of awareness programs on government schemes, scholarships and insurance.
- CHARUSAT Spoural Volunteer: participant coordination, registrations and event logistics.
- Languages: English, Gujarati, Hindi